

Prognostic and therapeutic implications of mTORC1 and Rictor expression in human breast cancer.



The mammalian target of rapamycin (mTOR) plays a key role in the regulation of cellular metabolism, growth and proliferation. It forms two multi-protein complexes known as complex 1 (mTORC1) and 2 (mTORC2). Raptor and Rictor are the core proteins for mTORC1 and mTORC2, respectively. This study examines the relationship between mTORC1, Rictor and Raptor mRNA expression and human breast cancer. Furthermore, the correlation between mTORC1 and hTERT was investigated. Breast cancer tissues (n=150) and normal tissues (n=31) were analysed using reverse transcription and quantitative PCR. Transcript levels were correlated with clinicopathological data. Higher mTOR expression was noted in breast cancer tissue (P=0.0018), higher grade tumours (grade 2 vs. 3, P=0.047), in ductal tumours (P=0.0014), and was associated with worse overall survival (P=0.01). Rictor expression was significantly higher in background breast tissues compared with tumours and was inversely related to the Nottingham Prognostic Index (NPI1 vs. 2, P=0.03) and tumour grade (grade 1 vs. 3, P=0.01) and was associated with better overall (P=0.037) and disease-free survival (P=0.048). The mRNA expression of Raptor was higher in tumours compared with normal tissues. Furthermore, the expression of Raptor was associated with a higher tumour grade (grade 1 vs. 3, P=0.027). A highly significant positive correlation between mTOR and hTERT (P<0.00001) was observed. These observations are consistent with the role of mTORC1 in the anti-apoptosis pathway and suggest that selective inhibitors of mTORC1 may be more efficacious in human breast cancer. Our findings support the hypothesis that mTORC1 is an important upregulator of telomerase in breast cancer.